

# INTELLIGENT AIRPORT CROWD MANAGEMENT SYSTEM

*Simulation-Based Crowd Monitoring, Prediction and Decision Support*

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## 1. Dataset Description

The prototype uses simulation-generated passenger-flow data rather than a fixed public dataset. The simulator continuously generates arrivals using a noisy time-of-day pattern and can inject a flight-surge event. Each airport zone records queue size, capacity, service rate, occupancy, throughput, and recent occupancy history. The five monitored zones are Entrance, Check-in, Security, Immigration, and Gate A12. The generated records are already numeric and require no missing-value or categorical encoding step; preprocessing consists of scaling time progression, calculating occupancy percentage, maintaining rolling history, and applying limits to predicted demand.

## 2. Problem Statement

Airports can experience rapidly changing passenger density, especially near check-in and security. Static monitoring may detect congestion only after queues have become severe. The objective of this project is to build an intelligent simulation that monitors current crowd density, predicts near-future demand, identifies critical zones, and recommends operational actions such as opening lanes, redirecting passengers, or reallocating staff.

## 3. Methodology

The system combines a browser-based queueing simulation with an AI Decision Engine. Passenger flow moves through the pipeline Entrance → Check-in → Security → Immigration → Gate A12. Each zone behaves as a queue with a capacity and service rate. Processed passengers become arrivals for the next zone, allowing congestion to propagate naturally. Current occupancy is classified as LOW, MEDIUM, or CRITICAL. A 30-minute demand prediction is produced using least-squares linear regression over recent occupancy history. When occupancy reaches approximately 80% or higher, the decision engine triggers an operational response: Open Lane (temporary service-rate increase), Staff Allocation (service-rate boost), or Redirect People (diversion of incoming passengers). A cooldown prevents repeated actions.

## 4. Results & Insights

The visual dashboard provides a live terminal-flow map, animated passenger movement, status indicators, a priority-zone card, a trend sparkline, and an action log. The monitor displays metrics such as current queue/capacity, average waiting time, and predicted crowd level. In a representative simulated state, Security can reach 187/200 passengers with an average waiting time of about 23 minutes, while the 30-minute projection can rise to roughly 245 passengers. During a surge, the model demonstrates how upstream flow increases downstream demand and how automated interventions change service capacity or incoming flow. These results validate the intended closed loop: current/future situation → decision → simulation → management action.

## 5. Novelty

The key innovation is the integration of real-time-style crowd monitoring, short-term prediction, and automated management actions inside one interactive simulation. Instead of showing only a static congestion chart, the prototype creates a feedback loop in which congestion changes the decision, and the decision changes the simulated passenger flow. The architecture is also extensible: a future implementation can replace synthetic arrivals with CCTV/YOLO detections and flight/passenger datasets, while the simulation layer can be connected to SUMO/TraCI, Mesa, or a dedicated pedestrian simulator.

## 6. System Architecture

Airport Side (CCTV/Computer Vision/YOLO) + Flight Side (Passenger and Flight Data/ML Prediction) → AI Decision Engine → Crowd/Passenger Simulation → Congestion Map → Management Action. The current prototype implements the behavior of this architecture in a self-contained browser simulation.